8.03SC Problem Set 4

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Problem 4.1

In the beaded string shown in Figure 1, the interval between neighboring beads is a, and the distance from the end beads to the wall is $a/2$. All the beads have mass m and are constrained to move only vertically in the plane of the paper. The strings are massless with constant string tension T.

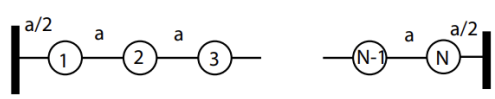

Figure 1: Beads on a string

a. Show that the physics of the left-hand wall can be incorporated by going to an infinite system and requiring the boundary condition $A_1 = -A_0$.

b. Find the analogous boundary condition for the right-hand wall.

c. Find the normal modes and the corresponding frequencies for the finite system.

a.

For small transverse displacements $y_n(t)$, the vertical force on bead n is

$$F_n = \frac{T}{a}(y_{n+1} - 2y_n + y_{n-1})$$

Thus the equation of motion is

$$m\ddot{y}_n = \frac{T}{a}(y_{n+1} - 2y_n + y_{n-1})$$

To incorporate the left wall, introduce a fictitious bead at $n = 0$. Since the wall is located halfway between beads 0 and 1 and is fixed, the displacement at the wall must vanish:

$$\frac{y_0 + y_1}{2} = 0$$

Hence,

$$\boxed{y_0 = -y_1}$$

In normal mode form $y_n(t) = A_n e^{i\omega t}$ this gives

$$\boxed{A_0 = -A_1}$$

b.

Similarly, introduce a fictitious bead at $n = N + 1$. The right wall lies midway between beads N and $N + 1$, so

$$\frac{y_N + y_{N+1}}{2} = 0$$

Thus,

$$\boxed{y_{N+1} = -y_N}$$

or in normal mode form,

$$\boxed{A_{N+1} = -A_N}$$

c.

Assume normal mode solutions

$$y_n(t) = A_n e^{i\omega t}$$

Substituting into the equation of motion gives

$$-m\omega^2 A_n = \frac{T}{a}(A_{n+1} - 2A_n + A_{n-1})$$

Rewriting,

$$A_{n+1} + A_{n-1} = \left(2 - \frac{ma\omega^2}{T}\right) A_n$$

Let

$$\lambda = 2 - \frac{ma\omega^2}{T}$$

The standard solution of this difference equation is

$$A_n = C\sin(kn) + D\cos(kn)$$

with

$$\lambda = 2\cos k$$

Therefore,

$$2\cos k = 2 - \frac{ma\omega^2}{T}$$

Hence,

$$\boxed{\omega^2 = \frac{2T}{ma}(1 - \cos k)}$$

Applying the boundary conditions yields

$$\sin(kN) = 0$$

so the allowed wave numbers are

$$\boxed{k_m = \frac{m\pi}{N} \qquad m = 1, 2, \ldots, N}$$

Thus the normal mode shapes are

$$\boxed{A_n^{(m)} = \sin\left(\frac{m\pi n}{N}\right) \qquad m = 1, 2, \ldots, N}$$

and the corresponding frequencies are

$$\boxed{\omega_m^2 = \frac{2T}{ma}\left(1 - \cos\left(\frac{m\pi}{N}\right)\right) \qquad m = 1, 2, \ldots, N}$$

There are exactly N normal modes.

Problem 4.4

A string of mass density μ and stretched with tension T is deformed as shown in the Figure 3 (amplitude A is very small, the deformation as shown in the figure is greatly exaggerated). The string is released at $t = 0$ with zero initial velocity ($\frac{\partial y(x,t)}{\partial t} = 0$ at $t = 0$ for all x).

a. How many normal modes do we have in this system?

b. Find expressions for the Fourier coefficients A_m of the normal mode expansion of the initial string shape and use them to write a full time-depenent series that describes motion of the string at $t > 0$: $y(x, t)$.

c. Make a sketch of the amplitudes A_m of the modes as a function of mode number m. Make sure that you indicate amplitudes for all possible normal modes of the string.

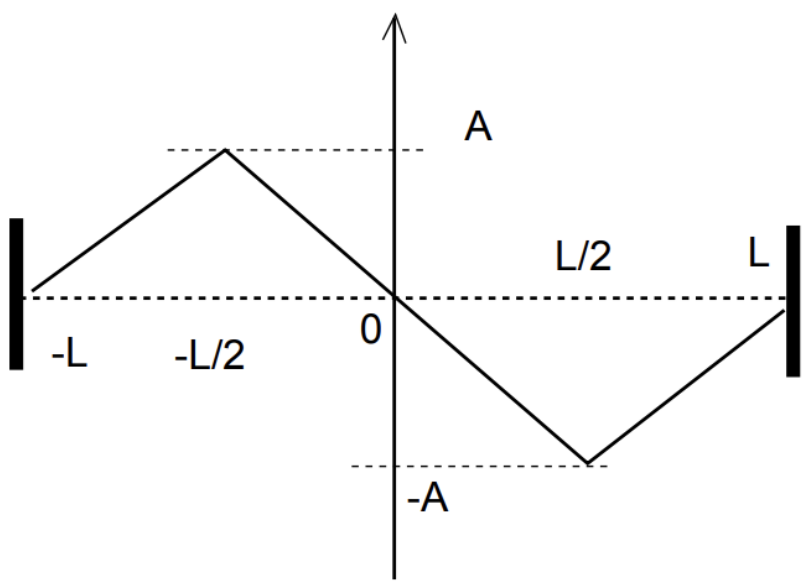

Figure 3: String Deformation

a.

The equation of Displacement of a standing wave is given by

$$y(x,t) = X(x)\cos(\omega t)$$

$$\implies \frac{\partial^2 y}{\partial t^2} = -\omega^2 \cos(\omega t) X(x) \qquad \frac{\partial^2 y}{\partial x^2} = \cos(\omega t) X''(x)$$

From the Wave equation,

$$\frac{\partial^2 y}{\partial t^2} = \frac{T}{\mu}\frac{\partial^2 y}{\partial x^2}$$

$$\implies \frac{T}{\mu}\cos(\omega t)X''(x) = -\omega^2\cos(\omega t)X(x)$$

$$\implies X(x) = A\sin(kx) + B\cos(kx)$$

By Boundary Conditions,

$$X(0) = 0 \implies B = 0$$

$$X(L) = 0 \implies A\sin(kL) = 0$$

For Non-Trivial Solutions to exist

$$\sin(kL) = 0 \implies k = \frac{m\pi}{L} \qquad m \in \mathbf{N}$$

Now,

$$\omega = kv$$

Therefore,

$$\omega_m = \frac{m\pi}{L}\sqrt{\frac{T}{\mu}}$$

So Normal Modes are given by

$$\boxed{y_m = \sin\left(\frac{m\pi}{L}x\right)\cos\left(\frac{m\pi}{L}\sqrt{\frac{T}{\mu}}t\right)} \qquad m \in \mathbf{N}$$

Hence, there are Infinite number of Normal Modes

b.

The string is fixed at both ends, so the normal modes are

$$\sin\left(\frac{m\pi x}{L}\right), \qquad m = 1, 2, 3, \ldots$$

with angular frequencies

$$\omega_m = \frac{m\pi}{L}\sqrt{\frac{T}{\mu}}.$$

Since the initial velocity is zero, the general solution can be written as

$$y(x,t) = \sum_{m=1}^{\infty} A_m \sin\left(\frac{m\pi x}{L}\right)\cos(\omega_m t)$$

At $t = 0$ we have

$$y(x,0) = \sum_{m=1}^{\infty} A_m \sin\left(\frac{m\pi x}{L}\right)$$

Thus the coefficients A_m are obtained by Fourier sine expansion of the initial shape:

$$A_m = \frac{2}{L}\int_0^L y(x,0)\sin\left(\frac{m\pi x}{L}\right)dx$$

If the initial shape is the triangular function

$$y(x,0) = \begin{cases} \dfrac{2A}{L}x, & 0 \le x \le \frac{L}{2}, \\ \dfrac{2A}{L}(L-x), & \frac{L}{2} \le x \le L, \end{cases}$$

then

$$A_m = \frac{2}{L}\left[\int_0^{L/2} \frac{2A}{L}x\sin\left(\frac{m\pi x}{L}\right)dx + \int_{L/2}^{L} \frac{2A}{L}(L-x)\sin\left(\frac{m\pi x}{L}\right)dx\right]$$

Evaluating the integrals gives

$$A_m = \frac{8A}{m^2\pi^2}\sin\left(\frac{m\pi}{2}\right)$$

Hence the full time-dependent solution is

$$\boxed{y(x,t) = \sum_{m=1}^{\infty} \frac{8A}{m^2\pi^2}\sin\left(\frac{m\pi}{2}\right)\sin\left(\frac{m\pi x}{L}\right)\cos\left(\frac{m\pi}{L}\sqrt{\frac{T}{\mu}}\,t\right)}$$

Only odd values of m contribute, since $\sin\left(\frac{m\pi}{2}\right) = 0$ for even m.

c.

From part (b), the Fourier coefficients are

$$A_m = \frac{8A}{m^2\pi^2}\sin\left(\frac{m\pi}{2}\right).$$

Since

$$\sin\left(\frac{m\pi}{2}\right) = \begin{cases} 0, & m \text{ even}, \\ \pm 1, & m \text{ odd}, \end{cases}$$

only odd modes contribute to the motion.
Thus,

$$A_m = \begin{cases} \dfrac{8A}{m^2\pi^2}(-1)^{\frac{m-1}{2}}, & m \text{ odd}, \\ 0, & m \text{ even}. \end{cases}$$

Therefore:

- The amplitudes vanish for all even m.
- The amplitudes decrease as $1/m^2$ for odd m.

Sketch description:
If one plots A_m versus m:

- Points exist only at $m = 1, 3, 5, \ldots$
- Their magnitudes decrease proportionally to $1/m^2$
- The sign alternates between successive odd values

Hence the amplitude spectrum consists of odd harmonics with rapidly decreasing magnitude.